The Selling closure: a complete operator set for reindexing, and the price of pseudo-symmetry

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Abstract

A monoclinic crystal measured twice can reduce once to a cell on c and once to a cell on $a+c$; the integer matrix between the two is not a lattice symmetry. Which integer matrices can relate two reduced cells, and how large must a search be to find them? Every matrix between two Selling-reduced cells of any lattice lies in one fixed set of $6096 = 48 \times 127$ matrices with entries up to ± 3. For a measured cell the set of reduced cells within a tolerance is finite and is enumerated exactly; below a radius computed for each cell, which exceeded the default tolerance for every PDB cell sampled, it reduces to a lookup of those 6096 matrices, a list smaller than the $\{-1, 0, 1\}$ window and, unlike it, with a stated range. A $\{-1, 0, 1\}$ window between Selling-reduced cells fails when a near-zero conorm of the reference lands on the fourth superbase vector, which reduction produces from a Niggli cell with three acute angles or from an arbitrary basis. A pseudo-twofold is priced at exactly twice its distance to the lattice it would make exact, about 5% strain per degree of Le Page obliquity. In the PDB the strain

between two deposited crystals of one form has a 90th percentile of 3–6%, depending on how a form is delimited; 21% of the 206 172 entries have a higher lattice symmetry within 3°, mostly by chance, and for 8–13% the cheapest alternative indexing from a higher lattice lies within that strain, so the cell alone cannot exclude it, although it ranks the wrong indexing first for only about one same-form pair in eighty.

1. Introduction

Take a primitive monoclinic lattice with $a = 30$, $b = 40$, $c = 50\,\text{Å}$ and $\beta = 100°$. Its Selling-reduced cell has two right angles between vectors of the reduced superbase, and a measured cell lies slightly to one side or the other of each. One data set therefore reduces to the cell 30, 40, 50 Å, $\beta = 100°$, and another, equally correctly, to 30, 40, 53.7 Å, $\beta = 113.4°$, built on a and $a + c$ (Section 3.2). The matrix between the two is not a symmetry of the lattice. A program that reports the Niggli cell meets the same multiplicity in another form: the cell on a and $a + c$ is never a Niggli cell, but at each right angle the Niggli cell changes between its two types, with three acute or no acute angles (Gruber, 1973; Andrews *et al.*, 1980), and two measurements of one crystal can be reported in Niggli cells related by a matrix that is not a symmetry either. Data-processing programs find such matrices by trying a list: small integer matrices between the two reduced cells, the matrices that unit-cell similarity searches generate (Grosse-Kunstleve *et al.*, 2002), or the operations of the lattice group found by a symmetry search (Le Page, 1982; Sauter *et al.*, 2004). On ordinary cases these searches succeed, and the question this paper answers is when they are guaranteed to: which integer matrices can relate two reduced cells, how large a search has to be, and how to price the pseudo-symmetric alternatives that a search returns.

The multiplicity has a classical description. A superbase is four lattice vectors summing to zero; it is obtuse, or Selling-reduced, when all six pairwise dot products are

non-positive (Selling, 1874). A superbase is obtuse exactly when its partial sums are Voronoi vectors of the lattice (Voronoi, 1908; Conway & Sloane, 1992), and the obtuse superbase is unique except when a dot product vanishes (Delaunay, 1933; Patterson & Love, 1957). Kurlin (2022) listed all obtuse superbases of each Voronoi type and derived from them a complete isometry invariant. Delone's 24 sorts classify lattices by the zeros and equalities among the six dot products (Delone *et al.*, 1975; Burzlaff & Zimmermann, 2006; Andrews & Bernstein, 2023). The S^6 representation (Andrews *et al.*, 2019*b*) writes the same six numbers as a point of a six-dimensional space, with boundary transformations and reflections between the reduced presentations of a lattice. On it rest the distances CS6Dist (Andrews *et al.*, 2019*b*) and NCDist (Andrews & Bernstein, 2014), the Bravais-type determination of SELLA (Andrews *et al.*, 2023), which projects onto the sorts' subspaces as Andrews & Bernstein (1988) projected onto the 44 Niggli characters in G^6, and the lattice matcher of LatticeRepLib (Andrews & Bernstein, 2026), which finds the presentation of one cell nearest to another by a fixed tree of boundary transformations and reflections. That a boundary transformation at a zero component carries one reduced presentation to another is noted in the SELLA paper (Andrews *et al.*, 2023, Section 6.3), and LatticeRepLib enumerates the reduced presentations reachable this way (`ExpandAcrossBoundaryZeros`); the set of these presentations is the closure of Section 3, and its size for each Voronoi type is Kurlin's (2022). Oishi-Tomiyasu (2012, 2024) gives error-stable Bravais-lattice determination in Selling coordinates with a provably sufficient finite list of basis transforms. The instability of the Niggli cell under error (Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) is the same multiplicity seen from the Niggli side, and the operators between two settings of one lattice form a coset of its symmetries (Flack, 1987).

This paper adds three things.

1. *A complete operator set.* Every integer matrix between two Selling-reduced cells

of any lattice lies in one set $\mathcal{U}$ of $6096 = 48 \times 127$ matrices, with entries up to ± 3; the proof is a short lemma and an exhaustive, computer-assisted enumeration (Section 3.3, Lemma A1). The entries reach 2 and 3 only when a near-zero conorm of the first cell sits on a slot of $d = -a - b - c$; a relabelling removes the 3s but not every 2 (Section 5.2).

2. *The tolerance under which it stays complete.* For a measured cell, the set of superbases that are reduced for some metric within a strain ϵ is finite and is enumerated exactly (Lemma A6). Below an adjacency radius computed for each cell, the set is a filter of $\mathcal{U}$ applied to the reduced cell, so a lookup of the 6096 matrices between two Selling-reduced cells is a complete search (Remark A7); the radius exceeds the default tolerance for each of 2000 PDB cells sampled. Reindexing a new cell against a reference widened this way returns every proper operator within the tolerance (Lemma A8).

3. *The price of pseudo-symmetry, calibrated on the PDB.* A candidate is priced by the affine-invariant distance between the two metrics. A pseudo-twofold costs exactly twice its distance to the lattice it would make exact (Proposition A9), about 5% per degree of Le Page obliquity. The strain between deposited cells of one crystal form sets the scale of the tolerance, and comparing it with the prices of the pseudo-symmetries in the PDB bounds how often the cell alone cannot decide the indexing (Section 5.5).

The guarantee matters where the searches in use have no stated range. On every case tested here the cctbx similarity searches, and a $\{-1, 0, 1\}$ search between the two Niggli cells, find the true operator as the closure does; a $\{-1, 0, 1\}$ search between the two Selling-reduced cells fails in the configurations that item 1 identifies. None of these searches says in advance for which cells it is complete; the lookup of $\mathcal{U}$ does, for

each cell.

Section 2 collects the classical definitions. Section 3 defines the closure, works through the monoclinic cell and defines $\mathcal{U}$. Section 4 gives the methods for measured cells, the pricing of pseudo-symmetry and the design of the tests, and states what is proved. Section 5 gives the results. Proofs are in Appendix A. Appendix B describes the distances of a measured cell to the 24 sorts, a by-product of the same computation. The supporting information holds catalogues, the construction of $\mathcal{U}$ with no lattice, the theory of an absolute band on the conorms, further tables and a list of the computational checks keyed to the sections of this paper.

2. Background

2.1. Superbases, conorms and the metric tensor

Selling reduction works with four vectors: the cell vectors a, b, c and $d = -a-b-c$. The four form a *superbase*, and any three of them are a basis of the lattice. The six negated pair products

$$p_{da} = -d\cdot a, \quad p_{db} = -d\cdot b, \quad p_{dc} = -d\cdot c, \quad p_{ab} = -a\cdot b, \quad p_{ac} = -a\cdot c, \quad p_{bc} = -b\cdot c$$

are the *conorms* (Selling, 1874; Conway & Sloane, 1992). Conorm vectors are always written in this slot order, da, db, dc, ab, ac, bc; the Selling parameters of S^6 are the same numbers with the opposite sign, in another order. A superbase is *obtuse*, which is to say Selling-reduced, when all six conorms are non-negative. The metric tensor in the basis (a, b, c) is linear in the conorms,

$$\mathbf{G} = \begin{pmatrix} p_{ab}+p_{ac}+p_{da} & -p_{ab} & -p_{ac} \\ -p_{ab} & p_{ab}+p_{bc}+p_{db} & -p_{bc} \\ -p_{ac} & -p_{bc} & p_{ac}+p_{bc}+p_{dc} \end{pmatrix}, \tag{1}$$

so the conorms are six cell parameters. A zero conorm is a right angle between two of the four vectors; the set of slots that vanish is the *zero pattern*, and the set of conorm vectors with $p \geq 0$ and those slots zero is a *face* of the cone of non-negative

conorms. A face describes lattices only if its non-zero slots connect the four vectors. Up to relabelling there are six kinds of face: no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector, and three zeros forming a triangle or a path. There are 38 faces in all. The triangle and the path describe the same lattices, those with three mutually perpendicular lattice vectors, read in different cells, so the six kinds of face are the five Voronoi types (Voronoi, 1908; Conway & Sloane, 1992). A *frame* is a choice of basis in which a metric tensor or a superbase is written.

2.2. Flips and swaps

Two kinds of integer operator act on a superbase.

Flips. A flip at the conorm p_{ab} replaces a by $-a$ and adds the old a to the two other vectors, $(a, b, c, d) \mapsto (-a,\ b,\ c + a,\ d + a)$. The new conorms are $p'_{ab} = -p_{ab}$, $p'_{ac} = p_{da} + p_{ab}$, $p'_{da} = p_{ac} + p_{ab}$, $p'_{bc} = p_{bc} + p_{ab}$, $p'_{db} = p_{db} + p_{ab}$ and $p'_{dc} = p_{dc} - p_{ab}$. Selling reduction applies flips at negative conorms until none remain. When p_{ab} is exactly zero the flip exchanges the slots ac and da and carries one obtuse superbase to another of the same lattice, which is why the reduced cell of a lattice with a right angle is not unique. In the basis (a, b, c) each flip is a fixed 3×3 integer matrix of determinant -1 that is its own inverse (Table 1). A word of flips such as $ab.bc.db$ is applied left to right, each flip being read in the labelling current at that step, and its matrix is $\mathbf{F}_{ab}\mathbf{F}_{bc}\mathbf{F}_{db}$, columns being the new basis vectors in the old basis.

Table 1. *The six Selling flips as* 3×3 *integer matrices in the basis* (a, b, c)*,* $d = -a - b - c$*; columns are the new basis vectors in the old basis. Each has determinant* -1 *and squares to the identity. For the flips at* da*,* db *and* dc *the vector* d *is negated and the new* a, b, c *are listed.*

flip at	new basis	matrix
ab	$(-a,\ b,\ a+c)$	$\begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
ac	$(-a,\ a+b,\ c)$	$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
bc	$(a+b,\ -b,\ c)$	$\begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
da	$(a,\ -a-c,\ -a-b)$	$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$
db	$(-b-c,\ b,\ -a-b)$	$\begin{pmatrix} 0 & 0 & -1 \\ -1 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$
dc	$(-b-c,\ -a-c,\ c)$	$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$

Swaps. A superbase is an unordered set of four vectors up to a common sign, so one reduced cell can be written in $4! \times 2 = 48$ ways. The 48 reorderings, with or without a change of the common sign, are the swaps. They change which vectors are called a, b, c and nothing else. Negating a single basis vector is not a swap: it does not preserve $a + b + c + d = 0$.

In S^6. A flip is a boundary transformation of S^6 (Andrews *et al.*, 2019*b*), written here on the basis rather than on the six parameters. A swap acts on the six Selling parameters as one of the 24 reflections of S^6; the common sign change acts trivially on dot products, so the 48 swaps give 24 reflections. The images of a cell under the boundary transformations and reflections that stay in the all-non-positive orthant of S^6 are its reduced presentations, the objects of the closure defined below.

2.3. Delone sorts and the root invariant

Delone (published as Delaunay, 1933) classified lattices into 24 sorts by the pattern of zeros and equalities among the reduced conorms. Each sort lies in one Bravais type, and some Bravais types comprise several sorts (Delone *et al.*, 1975; Burzlaff & Zimmermann, 1985, 2006). We use the symbols C1 to A3 and the patterns of Andrews & Bernstein (2023) and SELLA (Andrews *et al.*, 2023), read in the slot order of Section 2.1; the correspondence with their slot order is given in the supporting information. The sorts are the Selling counterpart of the 44 Niggli lattice characters (Niggli, 1928; de Wolff & Gruber, 1991). They split a Bravais type by the shape of the reduced cell, not by its symmetry. A zero conorm restricts the metric without adding a symmetry: one right angle between two superbase vectors gives no twofold, so the triclinic sorts A2 and A3, with one and two zeros, are as triclinic as A1.

All obtuse superbases of a lattice carry the same six conorm values as a multiset, since a flip at a zero conorm only exchanges two slots. Which slots hold which value is not invariant: in the monoclinic example the flip at ab moves p_{ac} into the slot da. Kurlin (2022) normalises the square roots of the conorms, by a rule that depends on the Voronoi type, to the root invariant, a complete isometry invariant of the lattice. The invariant identifies a lattice; the closure records how its reduced cells are related.

2.4. Terms

Table 2 collects the terms used in the rest of the paper, with the section in which each is defined. Strains and prices are given in per cent throughout; a band of 7.4% is 0.074 in the formulas.

Table 2. *Terms and symbols.*

term	meaning	defined
superbase, conorm	four lattice vectors summing to zero; the six negated pair products p_{ij}	2.1
obtuse, reduced cell	all six conorms ≥ 0; an obtuse superbase, in some labelling	2.1
zero pattern, face	the slots whose conorms vanish; the conorm vectors $p \geq 0$ with those zeros	2.1
frame	the basis in which a metric tensor or superbase is written	2.1
flip, swap	the operators of Table 1; the 48 relabellings of a superbase	2.2
sort	one of Delone's 24 classes of lattices by zeros and equalities among the conorms	2.3
visible symmetry, shape	a symmetry that relabels a reduced cell; the superbases whose conorms agree up to a swap	3.1
$\mathcal{U}$	the 6096 integer matrices between two reduced cells of some lattice	3.3
e	the empty flip word, the reduced cell itself	3.2
strain, $\|\cdot\|_{\mathbf{G}}$	$\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F$, equation (2)	4.1
geodesic distance Δ	the affine-invariant distance between two metrics, equation (3)	4.1
depth	the strain from a metric to the nearest metric that makes a superbase obtuse	4.2
price	the geodesic distance, after the best common scale, between a metric and the one an operator predicts	4.1, 4.2
band ϵ	the price within which an operator is accepted	4.4
widened closure $\mathcal{S}_\epsilon$	the superbases of depth at most ϵ	4.2
H_ϵ	the proper unimodular self-maps of a cell of price at most ϵ	4.4
parity class, μ_c	a class of lattice vectors modulo twice the lattice; the squared length of the shortest vector of class c	4.2
layer $\mathcal{N}_k$	$\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$	4.3
adjacency radius r_k	the strain at which a superbase outside $\mathcal{N}_k$ first becomes obtuse; computed up to 25%	4.3
difference-wall member	a member of the widened set reachable from the reduced cell only through a non-member	4.3
indexing class	a class of the Laue group (point group for anomalous data) among the candidates	4.4
obliquity δ	Le Page's angle between a lattice row and the nearest reciprocal row	4.5
non-isomorphism	the strain between two crystals of one form	4.5

3. The Selling closure

3.1. Definition

A lattice has no single preferred reduced cell. It has a finite set of obtuse superbases, and two kinds of integer matrix relate them: a symmetry maps a setting to itself and leaves its metric tensor unchanged, and a reindexing maps one setting to another, rewriting the metric tensor in the new basis without changing the lattice. A group holds only the first kind. A *groupoid* holds both: it is a collection of objects with

invertible arrows between them, in which two arrows can be composed only when the first ends where the second starts (Brandt, 1927; Higgins, 1971).

A *labelling* of an obtuse superbase is a choice of which three of its vectors, in which order and with which common sign, are called a, b, c; each obtuse superbase has 48 labellings, joined by swaps. Each labelling gives a basis and so a metric tensor. The *Selling closure* $\mathcal{C}(\mathbf{G})$ of a lattice has as objects these metric tensors, one for each labelling of each obtuse superbase, two labellings that a symmetry carries into each other giving the same tensor. Its arrows from $\mathbf{G}$ to $\mathbf{G}'$ are the unimodular matrices $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}'$, composed by matrix multiplication; it is the full subgroupoid, on these finitely many objects, of the action of $GL_3(\mathbb{Z})$ on positive definite matrices. The point group acts freely on labellings, since a symmetry that fixes a labelled basis is the identity, so a lattice with n obtuse superbases has $48n/|\text{point group}|$ objects. Elsewhere in the paper a *superbase* is the unlabelled set of four vectors, and the closure is counted by its n superbases. Three facts organise it.

1. *The arrows from an object to itself are the point group.* $\{\mathbf{P} : \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}\}$ is the lattice point group written in that basis (Himes & Mighell, 1987).

2. *The arrows between two objects are one coset of the point group.* If $\mathbf{P}_0$ is one arrow from $\mathbf{G}$ to $\mathbf{G}'$, every arrow between them is $g\mathbf{P}_0$ with g a symmetry of $\mathbf{G}$ (Flack, 1987).

3. *The arrows are known in advance.* A lattice has $n = 1$, 2, 3, 6 or 16 obtuse superbases according to its zero pattern (Kurlin, 2022; Lemma A1 gives the counts independently). Every arrow is a word of flips at zero conorms followed by a swap (Lemma A3), and every arrow lies in one fixed set $\mathcal{U}$ of integer matrices (Section 3.3).

Facts 1 and 2 hold for any connected groupoid. What is particular to the closure is

how the point group acts on the n superbases: the symmetries that merely relabel a superbase F_0 are its *visible* symmetries, the superbases in its orbit are those of the same *shape* (conorm vectors equal up to the slot permutation of a swap), and by the orbit–stabiliser theorem the order of the point group is the number of visible symmetries times the number of superbases of F_0's shape. Every symmetry that is not visible carries the reduced cell to a different reduced cell. A reduced cell here is always an obtuse superbase; a Niggli cell of type II (no acute angle) is one, and a Niggli cell of type I (three acute angles) is not.

3.2. One monoclinic cell

Take $a = 30$, $b = 40$, $c = 50\,\text{Å}$, $\beta = 100°$. Its reduced superbase is $M_0 = (d, a, b, c)$ with conorms

$$p_{da} = 639.5, \quad p_{db} = 1600.0, \quad p_{dc} = 2239.5, \quad p_{ab} = 0, \quad p_{ac} = 260.5, \quad p_{bc} = 0\ \text{Å}^2.$$

The two zeros are the right angles α and γ at the twofold axis b, present in every primitive monoclinic lattice. The closure has six superbases (Table 3). Each has two zero conorms, each flip at a zero leads to a neighbour, and the six form a ring M_0–M_1–M_5–M_4–M_3–M_2–M_0 (Fig. 1). They fall into three shapes. $\{M_0, M_4\}$ is 30, 40, 50 Å, $\beta = 100°$; $\{M_1, M_3\}$ is 30, 40, 53.7 Å, $\beta = 113.4°$, the cell on a and $a + c$; $\{M_2, M_5\}$ is the third cell choice, on c and $a + c$. These are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16).

Table 3. *The Selling closure of the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$: six obtuse superbases, each up to the order and common sign of its vectors, the flips at zero conorms that connect them, and where the four symmetries of the point group $2/m$ carry M_0.*

object	four vectors	flips lead to	symmetries carry M_0 here
M_0	$-a-b-c,\ a,\ b,\ c$	M_1, M_2	1, $\bar{1}$ (swaps of M_0)
M_1	$-b-c,\ -a,\ b,\ a+c$	M_0, M_5	
M_2	$-a-c,\ a+b,\ -b,\ c$	M_0, M_3	
M_3	$a+c,\ b-c,\ -b,\ -a$	M_2, M_4	
M_4	$a-b+c,\ -c,\ b,\ -a$	M_3, M_5	2 along b, m normal to b
M_5	$a-b,\ c,\ b,\ -a-c$	M_1, M_4	

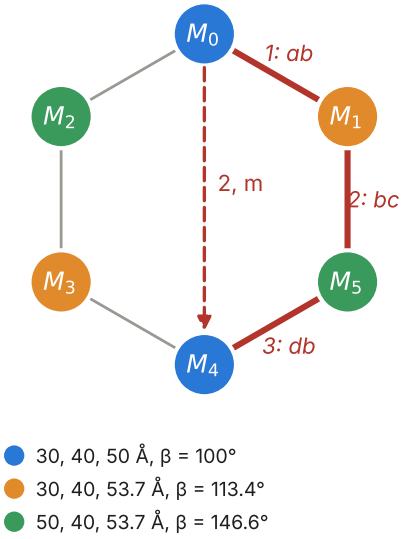


Fig. 1. The closure of the 100° cell: six superbases on a ring, joined by flips at the two zero conorms and coloured by cell shape. The red path is the flip word *ab.bc.db* from M_0 to M_4, each flip read in the labelling current at that step. The twofold and the mirror carry the vectors of M_0 to those of M_4 along this path, followed by a swap; they relabel no superbase on the way, so they are invisible in the reduced cell.

The point group is $2/m$, and the last column of Table 3 shows how its four operations act on the six superbases. The identity and the inversion are swaps of M_0: they relabel it. The twofold sends $(a, b, c) \mapsto (-a, b, -c)$ and so carries the four vectors $(-a - b - c, a, b, c)$ to $(a - b + c, -a, b, -c)$, which are the vectors of M_4, three flips away. As an arrow of the closure each symmetry is a loop at the metric tensor of M_0; as a word in flips and swaps the twofold is the flips *ab*, *bc*, *db* — the path $M_0 \to M_1 \to M_5 \to M_4$ — followed by the swap that renames the vectors of M_4 so that its metric tensor is that of M_0, and the mirror is the same flips with that swap composed with the change of common sign. Half of the point group is therefore invisible in the reduced cell: the stabiliser of M_0 has two elements, and the orbit of M_0 is $\{M_0, M_4\}$, as the count $4 = 2 \times 2$ requires. "Visible" means only that an operation relabels the reduced superbase; a search over lattice directions, such as Le Page's, finds the twofold either

241 way. Two measurements that reduce to M_0 and to M_1 report different cell parameters
242 and are related by no symmetry, only by the flip at ab.

243 *3.3. The universal operator set*

244 The arrows can be characterised with no lattice at all. An integer matrix $\mathbf{W}$ acts on
245 the six conorms by a fixed integer 6×6 matrix, and $\mathbf{W}$ is an arrow of the closure of
246 every lattice on a face exactly when that matrix maps the face into the cone of non-
247 negative conorms, a sign condition on its entries (supporting information, Section S3).
248 The union over the 38 faces is the set $\mathcal{U}$. It has $6096 = 48 \times 127$ elements with
249 entries in $\{-3, \ldots, 3\}$ and is closed under inversion. By Lemma A1 and the exhaustive
250 enumeration of its window, it contains every arrow between two reduced cells of any
251 lattice. The superbases $\mathcal{U}F_0$, 127 up to relabelling, are all the superbases that can be
252 obtuse together with F_0 for some metric. This set carries the completeness statements
253 of Section 4, and its entries say how large a search over integer matrices has to be
254 (Section 5.2).

255 ## 4. Measured cells: methods

256 A measured cell is never exactly on a zero conorm: a right angle is measured as 90.25°,
257 and the conorm is a small number of either sign. The flips that joined the superbases
258 of Table 3 now lead to superbases with a slightly negative conorm, Selling reduction
259 rejects them, and the exact closure of the measured metric has one superbase instead
260 of six. The closure has to be widened.

4.1. Two measures of mismatch

Two measures are used, both the same in every basis. The *strain* of a perturbation $\Delta\mathbf{G}$ of a metric $\mathbf{G}$ is

$$\|\Delta\mathbf{G}\|_{\mathbf{G}} = \|\mathbf{G}^{-1/2}\,\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F. \tag{2}$$

It is unchanged by a change of basis, since $\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}$ and $\Delta\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\Delta\mathbf{G}\mathbf{P}$ leave it as it is, and by a common scale of the cell. A perturbation of strain ϵ changes the dot product of any two lattice vectors by at most $\epsilon|u||v|$ (Proposition A4). The strain measures how far a metric must move to make a superbase obtuse, a convex problem in its six conorms. The *geodesic distance* between two metrics is

$$\Delta(\mathbf{G}_1, \mathbf{G}_2) = \Big(\sum_i \ln^2 \lambda_i\Big)^{1/2}, \tag{3}$$

with λ_i the eigenvalues of $\mathbf{G}_1^{-1/2}\mathbf{G}_2\mathbf{G}_1^{-1/2}$, the affine-invariant distance on positive definite matrices (Pennec *et al.*, 2006). It agrees with the strain to first order, it is a true distance (symmetric, with the triangle inequality), and every change of basis is an isometry for it. It measures how far apart two cells are, and it prices operators.

Table 4 converts the classical tolerances. For Le Page's twofold criterion (Le Page, 1982) the conversion is exact: a twofold whose obliquity, the angle between a lattice row and the nearest reciprocal row, is δ sits at a geodesic distance $\sqrt{2}\,\mathrm{artanh}(\sin\delta)$ from the metrics it would leave invariant, and its price is twice that (Proposition A9). One degree of obliquity is a price of 4.9%, and Le Page's 3° a price of 14.8%, for every twofold separately.

Table 4. *Classical tolerances in strain. The first two rows give the smallest strain of such a change, reached for an orthogonal pair (first row) or an axis normal to the other two (second row); the last two rows are exact for a twofold along a lattice row, in every cell (Proposition A9).*

tolerance	strain or price	value
angle between two vectors changed by δ, edges fixed	at least $\sqrt{2}\,\delta$	7.4% at 3°
edge changed by a fraction f, angles fixed	at least $2f$	10% at 5%
twofold of obliquity δ: geodesic distance to the metrics it fixes	$\sqrt{2}\,\operatorname{artanh}(\sin\delta)$	7.4% at 3°
twofold of obliquity δ: its price	$2\sqrt{2}\,\operatorname{artanh}(\sin\delta)$, 4.9% per degree	14.8% at 3°

4.2. The widened set

The *widened closure* of a metric $\mathbf{G}$ at strain ϵ is

$$\mathcal{S}_\epsilon(\mathbf{G}) = \left\{ F \;:\; F \text{ is obtuse for some metric within strain } \epsilon \text{ of } \mathbf{G} \right\}. \tag{4}$$

Its superbases are the reduced cells of every lattice consistent with the measurement at that tolerance. Each superbase has a *depth*, the strain from $\mathbf{G}$ to the nearest metric that makes it obtuse, and belongs to $\mathcal{S}_\epsilon(\mathbf{G})$ when its depth is at most ϵ. Equation (4) names no starting cell, order of operations or frame, so $\mathcal{S}_\epsilon(\mathbf{G})$ is the same whoever computes it. Each operator $\mathbf{P}$ from a metric $\mathbf{G}_1$ to a metric $\mathbf{G}_2$ has a *price*, the geodesic distance between $\mathbf{G}_2$ and $s\mathbf{P}^{\mathsf{T}}\mathbf{G}_1\mathbf{P}$ minimised over a common scale s, which removes the trace of the logarithm in equation (3). The price is zero for every arrow of an exact closure, and for a unimodular self-map the scale changes nothing.

It is finite. The lattice vectors modulo twice the lattice fall into eight *parity classes*, seven of which do not contain the zero vector; the seven partial sums of an obtuse superbase are shortest vectors of those classes. With μ_c the squared length of the shortest vector of class c, every vector of a superbase in $\mathcal{S}_\epsilon(\mathbf{G})$ lies in a ball of squared radius $\frac{1+\epsilon}{1-\epsilon}\mu_c$ (Lemma A6), so $\mathcal{S}_\epsilon(\mathbf{G})$ is enumerated exactly, with no window of coordinates to assume.

Widening one side suffices. If an operator $\mathbf{P}$ from a reference to a new cell has price at most ϵ, the new cell's reduced superbase, carried back by $\mathbf{P}$, lies in the reference's widened closure at strain $e^\epsilon - 1$ (Lemma A8). So only the reference is widened, and the new cell is reduced once.

4.3. Computing the widened set

For a measured cell and a strain $\epsilon < \frac{1}{4}$: Selling-reduce the cell to a superbase F_0; enumerate the lattice vectors in the seven balls of Lemma A6 and form the superbases they make; keep those of depth at most ϵ, with each one's depth and the operator $\mathbf{W}$ from F_0 to it. This is exact for every cell. Walking from the reduced cell, flipping wherever a conorm is within a band of zero, is not: a member can be reachable only through a non-member, when two small conorms lie at, say, 1.2 and 0.5 times the band and only the second of two flips returns into it. We call such a member a *difference-wall member*, since it lies across the wall where the two small conorms are equal; walks that miss such members are compared in Table S3 of the supporting information.

Layers and the lookup. Call two superbases adjacent when some metric makes both obtuse. The superbases adjacent to F_0 are the first layer $\mathcal{N}_1 = \mathcal{U}F_0$, and $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$. The *adjacency radius* $r_k(\mathbf{G})$ is the strain from $\mathbf{G}$ to the nearest metric for which a superbase outside $\mathcal{N}_k$ is obtuse; the implementation computes it up to 25%. If $\epsilon < r_1$, the widened set is the set of members of $\mathcal{N}_1$ of depth at most ϵ (Remark A7). It follows that, for two cells within price ϵ of each other with $e^\epsilon - 1 < r_1$ of the reference, trying the 6096 matrices of $\mathcal{U}$ between the two Selling-reduced cells finds every operator within ϵ. This lookup is a small-matrix search with a stated range: its 3048 proper matrices are fewer than the 3480 proper matrices with entries in $\{-1, 0, 1\}$. Here $\mathbf{W}$ is a column operator, $\mathbf{P} = B_R \mathbf{W} B_N^{-1}$; the implementation stores the elements of $\mathcal{U}$ as superbase rows, the transpose (`universal_operators` returns the column form).

Certifying the lookup takes one computation of r_1 for the reference, an enumeration of the same ball as the exact widened set; Section 5.3 reports it for the benchmark and for a sample of the PDB.

Ties take no band. Every sort lies on one zero pattern, and every symmetry of every lattice on that pattern is one of its arrows (Lemma A3), so a near-equality — the $a \approx b$ of a pseudo-tetragonal cell — never adds a superbase: the operator that would be its symmetry is already enumerated, and its price decides whether it is accepted.

4.4. Reindexing

A reference R is stored as its widened closure, the operators $\mathbf{W}_k$ from its reduced cell to the members. A new cell N is reduced once, to a basis B_N in its own coordinates; B_R is the reduction matrix of the reference. For each stored $\mathbf{W}_k$ and each of the 48 swaps $\mathbf{Q}$ the candidate operator from the reference's input basis to the new cell's is

$$\mathbf{P} = B_R \mathbf{W}_k \mathbf{Q}\, B_N^{-1}, \tag{5}$$

so that $(a_N\, b_N\, c_N) = (a_R\, b_R\, c_R)\mathbf{P}$, $\mathbf{G}_N \approx \mathbf{P}^{\mathsf{T}} \mathbf{G}_R \mathbf{P}$, and Miller indices transform as $h_N = \mathbf{P}^{\mathsf{T}} h_R$. Candidates of price at most the band ϵ are returned; the reference is widened at $e^{\epsilon} - 1$, so the true operator is among them whenever its price is at most ϵ (Lemma A8). Three conventions complete the procedure. Only operators of positive determinant are returned: cell parameters carry no hand, so the convention is that both bases are right-handed, and an inverted geometry must be caught before the cell reaches this step. A fitted scale that changes the edge lengths by more than 10% is rejected, and two primitive cells whose volumes differ by more than that allows, a factor 1.1^3, are refused with a message; a wrong centring symbol or a doubled axis lands here. An obverse–reverse relation between two rhombohedral lattices keeps the volume and the hexagonal cell, so no cell-based test sees it; it is a matter for the intensities. A centred reference is first taken to a primitive cell (a rhombohedral cell

on rhombohedral axes is already primitive) and the centring matrix composed with the operator found, which is then rational in the conventional basis.

Cosets and classes. Let H_ϵ be the proper unimodular self-maps of the reference of price at most ϵ, and p_0 the price of the true operator $\mathbf{P}_0$. Because the price is a distance, every candidate is $h\mathbf{P}_0$ with $h \in H_{\epsilon+p_0}$, and every $h \in H_{\epsilon-p_0}$ gives a candidate; if no self-map has a price in $(\epsilon, 2\epsilon]$, H_ϵ is a group, and if none has a price in $(\epsilon - p_0, \epsilon + p_0]$, the candidates are exactly the coset $H_\epsilon\mathbf{P}_0$ (Proposition A10). Two candidates that differ on the left by an element of the reference's Laue group $\mathcal{L}$ index the crystal the same way, so the distinct indexings are the classes of the proper part $\mathcal{L}^+$ of $\mathcal{L}$ among the candidates. For anomalous data from an acentric crystal the point group replaces the Laue group. The guarantee is that the true operator is among the candidates; the price orders the classes but does not pick the true member, which is a matter for the intensities.

4.5. Pricing pseudo-symmetry, and choosing the band

A symmetry g of a nearby lattice enters H_ϵ when its price is at most ϵ. For a twofold the price is exactly twice the geodesic distance from the measured metric to the metrics V_g that g leaves invariant; for a rotation of order n the price is at most twice that distance, and to first order near V_g the ratio is $\sqrt{3}$ for a threefold, between $\sqrt{2}$ and 2 for a fourfold and between 1 and $\sqrt{3}$ for a sixfold (Proposition A9). One threshold on the price of each operator therefore serves every sort, and Le Page's 3° is a price of 14.8% for each twofold.

The band is a tolerance on how much two measurements of one crystal may differ. For macromolecular crystals that difference is dominated by non-isomorphism, not by the measurement error that the standard uncertainties of a cell describe. We measure non-isomorphism in the PDB (Section 4.6) and take the scale of the band from its

upper percentiles, which depend on how a crystal form is delimited and on the Bravais type (Section 5.5). A candidate's price can then be read against that distribution: a class priced at the median non-isomorphism is as close to the reference as a typical second crystal of the same form, and the metric cannot exclude it. The tables of this paper use a band of 7.4%, the implementation's default, which lies above the pooled 95th percentile and the 90th percentile of every Bravais type except aP. For small-molecule data, whose cells are measured precisely and rarely non-isomorphous, a band of a few times the strain implied by the esds is used instead (`strain_from_esds` in the implementation).

4.6. Tests

Constructed cases. The benchmark (supporting information, Section S2) has 780 primitive cells with a random metric error and a random proper reindexing, built where the closure predicts trouble: 75 measured in another shape of the same lattice, 660 a small distance from a specialisation, 20 whose only reduced superbase is a difference-wall member, 16 needles and plates and 9 of the boundary sorts. Its errors are small (median strain 0.1%). The stress family puts independent errors on both cells of 60 references, scaled so that the true operator's price is 0.5 and 0.9 times the band. The window family probes the mechanisms of Section 5.2: 300 lattices with two right angles sharing a vector or three right angles, each measured twice with an error of 0.1 to 2%, the new cell in a random basis and the reference given either as its Niggli cell or in a random basis, and 60 lattices whose closure has a member needing an entry of 3, measured in that member. Eight methods are scored: the closure; the lookup of $\mathcal{U}$; searches over every matrix of determinant ± 1 with entries in $\{-1, 0, 1\}$ (3480 proper operators) and $\{-2, \ldots, 2\}$ between the two Selling-reduced cells; the same $\{-1, 0, 1\}$ search between the two Niggli cells; cctbx `similarity_transformations` between the

two Niggli cells (Grosse-Kunstleve *et al.*, 2002) with generator ranges 1 and 2; and a pipeline that runs the cctbx lattice-symmetry analysis (Sauter *et al.*, 2004) on both cells and returns every operation of the reference's highest lattice group between the two standard settings. Every method's candidates are priced the same way and kept within the band. On the window family the LatticeRepLib lattice matcher (`CmdLM`; Andrews & Bernstein, 2026) is run as well; it returns the presentation of the new cell nearest to the reference, not an operator, and is scored by whether that presentation lies within the band, a weaker criterion than the others'.

The PDB. The snapshot of 206 214 deposited cells (Zenodo record 22986222; see Data availability) is used twice. Non-isomorphism is measured between entries of one crystal form, taken as entries with the same space group whose identifiers are adjacent and share their first three characters, so that most pairs come from one deposition series; both cells are in the standard setting of the space group, so the identity is the operator, and pairs above 20% are counted as different forms. Pseudo-symmetry is measured by running the cctbx lattice-symmetry analysis at 3° on the primitive Niggli cell of every entry, recording the highest lattice group, its obliquity and the Delone sort of the idealised lattice, and pricing every proper operator of that group that is not a symmetry of the reported space group. Deposited cells are refined in the reported space group, so they satisfy its symmetry exactly; what the analysis measures is how far real crystals lie from a higher lattice, not measurement error.

4.7. What is proved

Table 5 separates what is proved from what rests on an exhaustive computation and what is checked. The proofs rest on Selling's identity, the characterisation of obtuse superbases by Voronoi vectors (Conway & Sloane, 1992; at a zero conorm a partial sum is a Voronoi vector that is not strict) and Selling reduction. Exact closures and

the closures of the 24 sorts were checked against brute force over bases with entries in $[-2, 2]$, a window that contains a labelling of every member of every exact closure (Lemma A1), and widened sets against the ball enumeration of Lemma A6.

Table 5. *What is proved, what rests on an exhaustive computation, and what is checked or refuted.*

statement	status
every arrow is flips at zero conorms and a swap	proved (Lemma A3)
the members of each closure lie in a finite window	proved (Lemma A1)
$\mathcal{U}$ contains every arrow between reduced cells of any lattice; the entry sizes by zero pattern	computer-assisted: the window enumerated over all 38 faces
the 24 sorts	a definition and an enumeration, matched to Delone's patterns by computation (Definition A5)
the strain bounds every dot product	proved (Proposition A4)
the widened set is finite and enumerated exactly	proved (Lemma A6)
below r_1 the widened set is a filter of $\mathcal{U}F_0$, and the lookup of $\mathcal{U}$ is complete	follows from the definition of r_1 and the completeness of $\mathcal{U}$ (Remark A7)
widening the reference alone suffices, at $e^{\epsilon} - 1$	proved (Lemma A8)
a twofold's price is twice its distance to the metrics it fixes, and $2\sqrt{2}\,\mathrm{artanh}(\sin\delta)$	proved, exactly (Proposition A9)
the ratios of price to distance for orders 3, 4, 6	proved in the strain at a symmetric metric; first order for the price (Proposition A9)
reduction from a type II Niggli cell keeps near-zero conorms off d where at most one of $a \cdot b$, $a \cdot c$, $b \cdot c$ is not small; from a type I cell near a three-zero face it puts them on d	proved (Section 5.2); from a type I cell near a two-zero face, observed
candidates lie between two cosets of self-maps; H_{ϵ} is a group in a gap of the prices	proved, exactly (Proposition A10)
the layers of 127 and 2755 superbases; the adjacency radii; the largest entries after relabelling	computer-assisted enumeration

5. Results

5.1. The closures of the 24 sorts

For each face the arrows are the operators of $\mathcal{U}$ that map the face into the cone; on the face every arrow permutes the six conorm slots, so each fixes a subspace of conorm vectors, and the sorts are the closed sets of equalities obtained by intersecting those subspaces (Definition A5). For a sort, the point group is the set of arrows that fix the sort's subspace pointwise, the visible symmetries are those that are swaps of the reduced cell, and two superbases have the same shape when their conorms agree up to a slot permutation induced by a swap. The resulting catalogue, with each sort's

pattern, number of superbases, symmetries, visible symmetries and shapes, is Table S9 of the supporting information; every entry is the result of this linear algebra on integer matrices. In every sort the number of symmetries out of each labelled superbase is the order of the holohedry, and the twofolds, with the inversion, generate the group.

Symmetry visible in one cell. In the sorts with zero conorms the symmetries that relabel the reduced cell are a minority: 2 of 8 for primitive orthorhombic, 4 of 16 for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal. None of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell, nor the one twofold of primitive monoclinic. The orbit of the reduced cell under the point group has |group|/visible superbases, all of its shape: for oP the orthogonal cell and three monoclinic-looking cells with two right angles, one for each axis traded for a face diagonal. Reduction of a measured orthorhombic cell stops at whichever of the sixteen superbases its errors favour; an orthogonal superbase is always one flip away, but the one in the reference's labelling can be four flips away, and no twofold of the lattice reaches it. The catalogues of shapes and of the conditions under which one sort becomes another are Tables S1 and S2 of the supporting information.

5.2. How large a search must be

Table 6 gives the largest entry of the operators of $\mathcal{U}$ out of a reduced cell, written in that cell's basis, for each kind of face. With no zero conorm, one zero, or two zeros on disjoint pairs, every operator has entries in $\{-1, 0, 1\}$. With two zeros sharing a vector or with three zeros, the entries reach 2 when a zero sits on a slot of $d = -a - b - c$, and 3 for a path of three zeros with two on d, such as conorms $(0, 0, 3, 2, 0, 5)$ with the obtuse basis $(-3a - 2b - c,\ a,\ a + b)$. A relabelling of the reduced cell that moves its zeros off d brings every operator into $\{-1, 0, 1\}$ wherever such a labelling exists: for one zero, two zeros sharing a vector and a triangle of three (two zeros on disjoint pairs

need only entries of 1 in any case). A path of three zeros touches all four vectors, so no labelling keeps it off d, and in the best of the 48 labellings an operator still needs an entry of 2 (`face_labellings.py`). Entries of 3 are therefore a property of the labelling that reduction returns; entries of 2 on a path are a property of the reduced cell.

Table 6. *The largest entry of an operator from a reduced cell to another reduced cell of the same lattice, in the basis of the first, by kind of face and by whether a zero conorm sits on a slot of d; over all 38 faces.*

zero conorms	zeros avoid d	a zero on d
none, or one	1	1
two on disjoint pairs (O3, cF, R3, M3, A3)	—	1
two sharing a vector (mP, oS, hP)	1	2
three, a triangle (oP, tP, cP from an orthogonal cell)	1	2
three, a path (the same lattices from a face-diagonal cell)	—	2 or 3

When does reduction put a zero on d? The entries are read in the basis of the first cell, so the question is about the reference's reduced superbase. When reduction starts from the Niggli cell, as it does when a program reports that cell, the Niggli type answers it, and for a lattice near primitive orthorhombic the reason is a sign. Let the Niggli cell be (a, b, c). In a type II cell (no acute angle, $a \cdot b$, $a \cdot c$, $b \cdot c \leq 0$) the conorms p_{ab}, p_{ac}, p_{bc} are non-negative, and Niggli's conditions $|2a \cdot b| \leq a \cdot a$, $|2a \cdot c| \leq a \cdot a$, $|2b \cdot c| \leq b \cdot b$ with $a \leq b \leq c$ make $p_{da} = a \cdot a + a \cdot b + a \cdot c \geq 0$ and likewise $p_{db}, p_{dc} \geq 0$, so the Niggli cell is itself obtuse. Near mP or oP at most one of the three products is not small, and then each conorm of d is at least half a squared edge less the small products, so the near-zero conorms are not on d. In a type I cell (all three products positive) the orthogonal cells $(\pm a, \pm b, \pm c)$ cannot be obtuse, since negating basis vectors does not change the sign of $(a \cdot b)(a \cdot c)(b \cdot c)$, which an obtuse orthogonal cell needs to be non-positive. Near a three-zero face the measured reference therefore reduces to a face-diagonal cell, whose zeros form a path through d (checked on 200 measured cells; supporting information, Table S10). Near a two-zero face the same holds less sharply and depends on the order of flips: a type II Niggli cell kept its right angles on ab, ac, bc in 519 of 519 measured near-mP cells, and from a type I cell

reduction put a near-zero conorm on a slot of d in 614 of 681, both of them in 316. Which type a measurement gets is decided by the signs of its small errors, so about half of all measurements of a near-monoclinic or near-orthorhombic lattice are type I.

The Niggli type is thus a property of how the reference is presented, not of the reference: given in another basis, a type II reference can also reduce with a zero on d (Section 5.4). What decides the entries is Table 6 applied to the reduced superbase that reduction returns.

5.3. Completeness on measured cells

The worked cell. The measured cell 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80° is the 100° cell of Section 3.2 with β moved to 92°, perturbed by a metric error of strain 0.82%. Its widened closure grows with the strain: one member at 0.1%, two at 0.3%, three at 0.6%, the six of the ring from 1%, and sixteen at 7.4% (Fig. 2; the members, as flip words with their conorms and depths, are in Table S6). The six members of the ring enter below 0.8%; the other ten members of the closure of the nearby orthorhombic lattice enter between 3.8 and 5.1%.

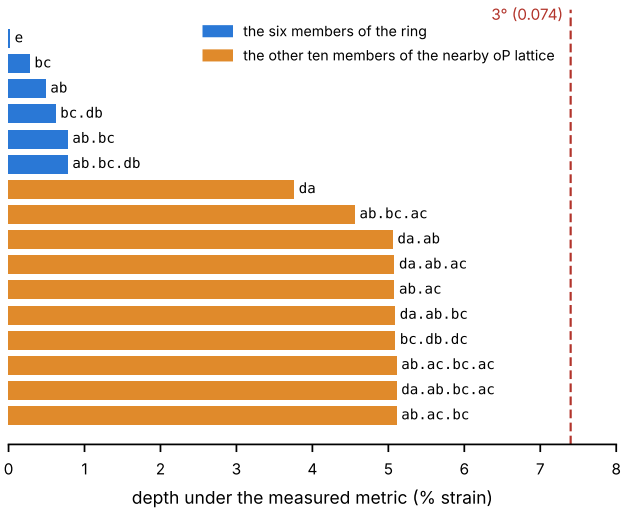


Fig. 2. The sixteen members of the widened closure of the measured worked cell by depth: the six of the ring enter below 0.8% strain, the other ten members of the nearby orthorhombic lattice's closure between 3.8 and 5.1%.

Needles and plates. The exact enumeration and brute force agree on a needle $5.58 \times 5.21 \times 131.97$ Å, $\gamma = 91.12°$, at strains from 0.05 to 5.2%, on a $100 \times 100.1 \times 5$ Å and a $150 \times 120 \times 4.8$ Å plate at 5.2%, where the widened sets have 20 and 28 members, and on 240 random needles, plates, isotropic and monoclinic cells. For very thin cells at large bands the ball of Lemma A6 grows quickly (10^4 lattice vectors for a $5 \times 5 \times 500$ Å needle at 7.7%); the implementation refuses balls above 5000 vectors with a message rather than run for minutes.

Adjacency radii. Table 7 gives the adjacency radii. For the worked cell and a $10 \times 10 \times 30$ Å needle no superbase outside the first layer comes within 25%, so the lookup of $\mathcal{U}$ is complete at every band in use. For needles and plates r_1 falls with the ratio of the shortest to the longest edge, to 4–6% for the thinnest cells, and the second layer roughly doubles it; there the lookup is complete only for bands below r_1, and the exact enumeration is needed beyond. For the BLDConograph sample cells and a lysozyme cell (Appendix B) r_1 reaches the cap, and so it does for 99.4% of the 697 benchmark

512 references and 98.6% of 2000 PDB cells drawn at random; every one of those PDB
513 cells has r_1 above the 7.7% that certifies the default band, and computing r_1 took a
514 median of 13 ms per cell. On a tenth of the benchmark, widening a reference took a
515 median of 0.02 s and reindexing one new cell 0.03 s (90th percentiles 0.04 and 0.17 s);
516 a cubic reference or a thin plate took up to 0.9 s to widen and 0.4 s per new cell in our
517 tests.

Table 7. *Adjacency radii in strain (%): r_1, the smallest strain at which a superbase outside the first layer becomes obtuse, and r_2 for the second layer, computed up to* 25%.

cell	r_1	r_2
$30 \times 40 \times 50$ Å, $\beta = 92°$ (the worked cell)	> 25	> 25
needle $10 \times 10 \times 30$ Å	> 25	> 25
needle $10 \times 10 \times 100$ Å	13.9	> 25
needle $5.58 \times 5.21 \times 132$ Å, $\gamma = 91.12°$	5.6	11.1
plate $100 \times 100.1 \times 5$ Å	5.0	14.1
plate $150 \times 120 \times 4.8$ Å, $\beta = 90.4°$	4.3	10.0

518

519 *5.4. Reindexing*

520 *Indexing classes.* Table 8 and Fig. 3 give the classes for six references against the
521 same crystal measured again with a 0.1% error in the length scale, which the scale
522 fit absorbs, and a relabelling. For the exactly symmetric references the counts are
523 the classical indexing ambiguities of merohedral crystals (Yeates, 1997; Dauter, 2003;
524 Evans, 2006, 2011): one class for $P4_32_12$, four for $P3$, two for $P3_121$ and $P2_13$. The
525 price adds the pseudo-merohedral classes: a $40.0 \times 40.3 \times 80$ Å $P2_12_12_1$ cell and the
526 worked cell in $P1$ each have a second class, at 2.11% and 1.55%, twice, to first order,
527 the cells' distances of 1.06% and 0.78% to the sorts the twofolds would make exact,
528 from the twofold that would make the cell tetragonal or monoclinic. With proper
529 operators only, anomalous data split no class of a chiral crystal; for an acentric crystal
530 that is not chiral the point group replaces the Laue group, and $P4mm$ goes from one
531 class to two and $P31m$ from two to four.

Table 8. *Distinct indexings of six references at a band of* 7.4% *(the reference widened at* 7.7%*): the order of the band's group of proper self-maps* H_ϵ *and of the proper rotations* $\mathcal{L}^+$ *of the Laue group, the number of classes among the candidates, and the price of each class. The new cell is the reference with a* 0.1% *scale error, which the scale fit absorbs, relabelled.*

reference	space group	$\lvert H_\epsilon\rvert$	$\lvert\mathcal{L}^+\rvert$	classes	prices (%)
$79.1 \times 79.1 \times 37.9$ Å	$P4_32_12$	8	8	1	0
$a = b = 50$, $c = 80$ Å	$P3$	12	3	4	0, 0, 0, 0
$a = b = 50$, $c = 80$ Å	$P3_121$	12	6	2	0, 0
$a = 60$ Å	$P2_13$	24	12	2	0, 0
$40.0 \times 40.3 \times 80$ Å	$P2_12_12_1$	8	4	2	0, 2.11
worked cell	$P1$	2	1	2	0, 1.55

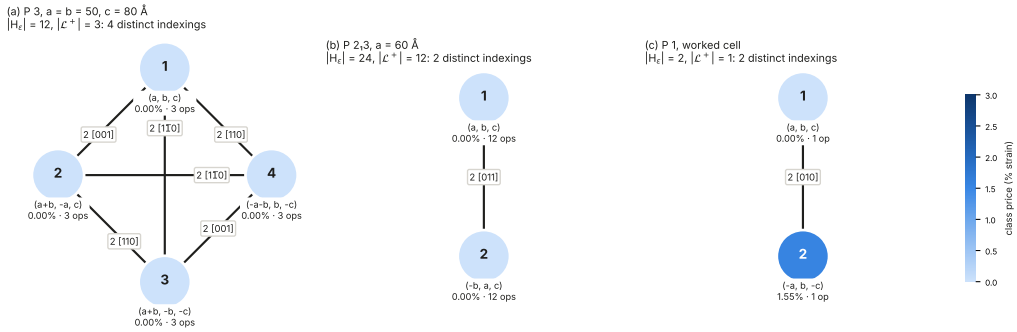


Fig. 3. Three of the cases of Table 8 as graphs: the classes among the candidates for (*a*) $P3$, (*b*) $P2_13$ and (*c*) the worked cell in $P1$. Each node is one class, with its operator nearest the identity (new basis in terms of the reference's a, b, c), its price and the number of operators in it; each edge is the lattice symmetry or pseudo-symmetry carrying one class to the other. Node colour is the price.

Methods compared. Table 9 scores the methods of Section 4.6 on the four families. The closure found the true operator in every case, as Lemma A8 guarantees, and so did the lookup of $\mathcal{U}$, including for the 12 benchmark references, all needles and plates, whose r_1 lies below the 7.7% that would certify it. So did the $\{-1, 0, 1\}$ search between the two Niggli cells and both cctbx similarity searches, in every family; the benchmark does not separate them from the closure. The $\{-2, \ldots, 2\}$ search between Selling-reduced cells missed one stress case, eight entry-3 cases and two near-orthogonal cases. The $\{-1, 0, 1\}$ search between Selling-reduced cells is the one that fails, and where Table 6 predicts: on the benchmark it missed two triclinic cells near mP whose type I references put a near-zero conorm on d and the operator's entry at 2; on the 60 cells

measured in a member that needs an entry of 3 it found 24. With the reference given as its Niggli cell it found every operator from the 129 type II references and missed 29 of 171 from type I references, 23 of 76 near the three-zero faces, where every type I reference has its zeros on d, and 6 of 95 near the two-zero faces. With the reference given in a random basis, so that reduction does not start from the Niggli cell, it missed 81 of 300, 44 of them from references whose Niggli cell is type II: the Niggli type predicts the failures only when reduction starts from the Niggli cell. The pipeline missed one benchmark needle, $5\times5\times100$ Å with $\beta = 92°$, whose reference and measured cell idealise at 3° to different lattices, ten stress cases and eight window cases. Per benchmark case the closure took 0.08 s including the widening of the reference, the $\{-2,\dots,2\}$ search 0.8 s and the other methods 0.04 s or less.

Table 9. *The true operator among the candidates, by method and family; candidates priced by the geodesic distance after the best scale and kept within a band of* 7.4% *(the reference widened at* 7.7% *for the closure and the lookup). For the lattice matcher, which returns a cell, a hit is a presentation within the band. Window family: cells measured in a member that needs an entry of* 3*, and near-mP and near-orthogonal lattices measured twice with an error of* 0.1 *to* 2%*, the reference given as its Niggli cell (split by Niggli type) or in a random basis. The lattice matcher is scored on a weaker criterion than the operator searches. A dash: not run. The lookup on the benchmark and stress families is in* `results_lookup.txt`*. Results in* `manuscript/benchmark/`*.*

					2- or 3-zero faces, reference as		
		stress			Niggli cell		random
method	benchmark	0.5ϵ	0.9ϵ	entry 3	type I	type II	basis
closure	780/780	60/60	60/60	60/60	171/171	129/129	300/300
lookup of $\mathcal{U}$	780/780	60/60	60/60	60/60	171/171	129/129	300/300
$\{-1,0,1\}$, Selling cells	778/780	58/60	52/60	24/60	142/171	129/129	219/300
$\{-2,\dots,2\}$, Selling cells	780/780	59/60	60/60	52/60	170/171	129/129	299/300
$\{-1,0,1\}$, Niggli cells	780/780	60/60	60/60	60/60	171/171	129/129	300/300
cctbx similarity, range 1	780/780	60/60	60/60	60/60	171/171	129/129	300/300
cctbx similarity, range 2	780/780	60/60	60/60	60/60	171/171	129/129	300/300
lattice-symmetry pipeline	779/780	56/60	54/60	60/60	171/171	126/129	295/300
lattice matcher (`CmdLM`)	—	—	—	60/60	171/171	129/129	300/300

What the metric does not decide. The cheapest candidate lies in the true operator's coset of the exact lattice symmetries in 742 of the 780 benchmark cases, and is the true operator itself in 741. All 20 difference-wall cases fail this test: there the cheapest candidate is the true operator composed with a pseudo-symmetry of the nearby

orthorhombic corner, which costs less than the true operator at that error. In the stress cases the cheapest candidate is in the true coset in 68 of 120, and is the true operator itself in 40. When a pseudo-symmetry costs less than the error, the price orders the candidates but cannot pick the true one; how often that happens in real crystals is the subject of the next section.

5.5. Pseudo-symmetry in the PDB

Non-isomorphism. Between deposited cells of one crystal form (45 222 pairs) the price after the scale fit has a median of 0.98%, a 90th percentile of 4.2% and a 95th of 6.7%; without the scale fit the median is 1.6%. The upper percentiles depend on how a crystal form is delimited. With the cut between forms at 5, 10, 20 and 50% the 95th percentile is 3.1, 4.7, 6.7 and 11.9%; dropping the 1 464 pairs of identical cells changes little (6.9%), and weighting each deposition block equally, so that fragment-screening series of up to 35 pairs count once, gives 5.0 and 8.0% for the 90th and 95th. They also depend on the Bravais type: the 95th percentile is 4.0% for tP, 4.6% for hP, 7.4% for oP, 9.0% for mP and 11.7% for aP, and 0 for the cubic types, whose shape the scale fit removes entirely (`results_pdb_band_controls.txt`). The scale a reindexing band has to cover for macromolecular data is therefore a few per cent, about 3–6% at the 90th percentile and 4–9% at the 95th for the non-cubic types, more for triclinic crystals; the standard uncertainties of a single cell describe its measurement error alone and do not reach it.

Pseudo-symmetry. Le Page's analysis at 3° finds a higher lattice symmetry than the reported one for 43 232 of the 206 172 entries analysed (21%); for 1 189 (0.6%) the extra symmetry is exact, the metric of a merohedral crystal. Most of the rest is what a tolerance of 3° admits by chance: shuffling the free cell parameters independently among the entries of one space group, which keeps their distributions and destroys any

relation between them, gives nearly the same rate for $P1$ (39.5 against 42.3%), $P2_1$ (44.8 against 44.4%), $C2$ and $P2_12_12$, and a clear excess only for $P2_12_12_1$ (15.2 against 22.6%), whose real cells are more often pseudo-tetragonal. The operators are those of the cctbx analysis, priced as in this paper. For the 42 043 entries with no exact extra symmetry the largest price matches $2\sqrt{2}\,\mathrm{artanh}(\sin\delta)$ for the group's largest obliquity δ (ratio 1.000 at the 10th, 50th and 90th percentiles), as Proposition A9 requires of a twofold, and the cheapest operator has a median price of 6.7%. Fig. 4 sets these prices against non-isomorphism: the cheapest alternative lies within the 90th percentile of non-isomorphism for 35% of the pseudo-symmetric entries and within the 95th for 50%. Over all entries, 7.8% have an alternative, exact or priced, within 4.2% and 10.8% within 6.7%; with the band of the entry's own Bravais type, 8.9 and 12.7%.

What the cell decides. An alternative within the band is one the cell does not exclude; it is not one the cell prefers. Reindexing the second cell of 3 000 randomly chosen same-form pairs against the first, with the first cell's space group and a band of 7.4%, a class other than the identity's is within the band at exactly the identity's price in 17% of pairs, an exact lattice symmetry outside the Laue group (the classical merohedral ambiguities within the reported lattice type, such as $P3$ or $P4$ in a lattice of higher holohedry, which Le Page's analysis does not count and no cell can resolve), and at a positive price in 7.9%. A wrong class is strictly cheaper than the identity's in 1.2% of pairs, and the identity lies beyond the band in 4.2%. So for about one pair in thirteen the cell leaves a priced alternative open, and for about one in eighty it ranks the wrong indexing first.

Where. The share depends on the reported type and on the transition (Table 10). Monoclinic and triclinic entries carry most of it: 18% of the 35 364 mP entries and 16% of the 8 522 aP entries have an alternative within 4.2% (exact ones included), against 7% of oP and 1% of tP entries; tP entries reported in their full symmetry

have little higher symmetry left to find. Among the transitions, mP to hP is nearly always undecidable from the cell (median price 0.71%, 91% within 4.2%): hexagonal metrics reported in a monoclinic group, the configuration in which pseudo-merohedral twinning is possible. The median price of the cheapest alternative is 6–7.5% for most transitions, so the sort decides how often a pseudo-symmetry occurs, not what it costs (supporting information, Table S8).

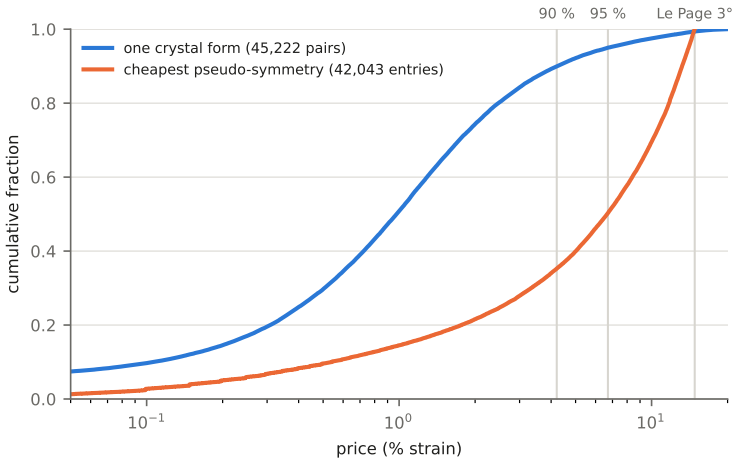


Fig. 4. Non-isomorphism against pseudo-symmetry in the PDB: cumulative distributions of the price between deposited cells of one crystal form (blue; one value per pair) and of the price of the cheapest pseudo-symmetric alternative of each entry for which Le Page's 3° finds a higher lattice that is not exact (orange; one value per entry). Vertical lines: the 90th and 95th percentiles of non-isomorphism, 4.2 and 6.7%, and the price of a twofold at Le Page's 3°, 14.8%.

Table 10. *The cheapest pseudo-symmetric alternative of PDB entries, by transition from the reported Bravais type to the one Le Page's* 3° *finds: entries with priced alternatives, the median price (%), and the share within* 1%*, within the 90th percentile of non-isomorphism (*4.2%*) and within its 95th (*6.7%*). The sixteen most frequent transitions.*

transition	entries	median	≤ 1%	≤ 4.2%	≤ 6.7%
oP→tP	12 120	6.98	9	31	48
mP→oS	9 474	7.38	12	33	46
mP→oP	4 417	5.78	20	42	55
aP→mC	2 113	9.49	9	19	30
mC→oS	1 831	7.41	12	33	45
mP→tP	1 542	3.66	27	54	69
mC→oI	1 308	7.13	11	30	48
oS→hP	1 169	7.34	10	26	45
oI→tI	1 165	6.25	19	38	52
mC→oF	1 067	7.29	8	29	46
oS→tP	880	7.42	14	32	46
tP→cP	620	7.06	7	27	47
aP→mP	571	6.97	15	30	48
mP→hP	479	0.71	61	91	100
mC→hR	391	8.92	19	29	37
mC→tI	391	4.56	23	46	64
all	42 043	6.67	15	35	50

6. Discussion

A guarantee where searches have none. On every case tested, the cctbx similarity searches and a $\{-1, 0, 1\}$ search between the two Niggli cells found the true operator, as the closure did; the benchmark shows no practical advantage of the closure over them. What the closure adds is a statement of range. The lookup of the 6096 matrices of $\mathcal{U}$ between the two Selling-reduced cells is complete below the adjacency radius of the reference, which is computed for each cell in milliseconds and exceeded the default tolerance for every PDB cell sampled; beyond it the exact enumeration takes over. A fixed window of entries, or the operations of one lattice group, carries no such statement. Between Selling-reduced cells a $\{-1, 0, 1\}$ window fails exactly where Table 6 says it can, when the reference's reduced superbase has a near-zero conorm on a slot of d, and it missed one operator in four when the reference was given in a random basis. That the same window between Niggli cells never failed is an observation on

these constructions, not a bound; no size of operator between the Niggli cells of a lattice and of a nearby measurement of it is known to us.

Pseudo-symmetry and non-isomorphism. The price of a pseudo-symmetric alternative is a geodesic distance and can be compared directly with the distance between two crystals of one form. Where the alternative costs less than ordinary non-isomorphism, no metric criterion can exclude it and the intensities must decide, as in twinning and indexing-ambiguity tests (Yeates, 1997; Evans, 2006, 2011; Brehm & Diederichs, 2014; Gildea & Winter, 2018). In the PDB that is the case for about 8–13% of entries, depending on the percentile and on whether the band is set per Bravais type, and for about one pair in thirteen of one crystal form; the cell ranks the wrong indexing first far less often, for about one pair in eighty. These are upper and lower views of the same question, and both rest on a proxy for "one crystal form" whose tail mixes in other crystals. Most pseudo-symmetry at 3° is what that tolerance admits by chance, so a pseudo-symmetric metric is not by itself evidence of twinning. The rule of thumb, about 5% of price per degree of Le Page obliquity, makes the two scales comparable: the 90th percentile of non-isomorphism is about 0.85°, and Le Page's 3°, a tolerance for assigning a Bravais lattice, is about three times wider than the band needed for reindexing. For twofolds one threshold per operator serves every Delone sort, since the price depends on the obliquity and not on the sort; the sort decides how often each kind of pseudo-symmetry occurs (Table S8).

The Niggli cell. The instability of the Niggli cell (Gruber, 1973; Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) is the same phenomenon in part. At a zero scalar product the Niggli cell changes between type I and type II, the two sign classes of Niggli's reduced form, and the closure handles that boundary as it handles any zero conorm. At a tie such as $a = b$, and at Niggli's special conditions, the Niggli basis changes while the Selling-reduced superbase does not: such changes are relabellings of

one superbase, as Andrews *et al.* (2019*a*) observed in comparing the two reductions. In the benchmark the measured cell's Niggli cell differs from the reference's in 45 of 780 cases and is the same cell in another basis in 125 more (supporting information, Section S2). Tolerant Niggli reduction (Křivý & Gruber, 1976; Grosse-Kunstleve *et al.*, 2004) returns one of these cells, not the set.

Relation to S^6, SELLA, the lattice matcher and BLDConograph. The flips and swaps are the boundary transformations and reflections of S^6 (Andrews *et al.*, 2019*b*), and the closure of a lattice is the set of its images under those maps that stay in the reduced orthant, the set that LatticeRepLib's `ExpandAcrossBoundaryZeros` generates. The LatticeRepLib lattice matcher searches a fixed tree of these maps, of depth two, for the presentation of a cell nearest to a reference: a fixed list of words, as $\mathcal{U}$ is a fixed list of matrices. It found such a presentation in every window case (Section 5.4), on a weaker criterion than the operator searches, and returns a cell rather than the operator and the classes; whether its tree contains $\mathcal{U}$ we have not checked, and it carries no statement of range. CS6Dist and NCDist are metrics on lattice space, suitable for database search, and return a distance, not an operator. Oishi-Tomiyasu (2012, 2024) bounds the basis transforms needed for Bravais-lattice determination under an error bound; the layers and adjacency radii here do the same for reindexing, with the radius computed for each cell. Appendix B compares the distances of a cell to the 24 sorts with SELLA.

Scope. "Complete" in this paper means: every proper unimodular operator whose price is at most the band is among the candidates, for a band the user supplies. The enumeration is exact below a strain of 25%, and the lookup of $\mathcal{U}$ below the adjacency radius. Sub- and superlattice relations (a doubled axis, determinant ± 2 or more) are outside the closure and are refused with a message; an obverse–reverse relation between two rhombohedral lattices keeps the cell and is not seen at all; the choice

within a class is left to the intensities. The PDB measurements describe deposited cells, refined with their reported symmetry imposed, and the non-isomorphism proxy, adjacent identifiers of one space group, mixes in different proteins and crystal forms at the tail, so its upper percentiles are upper estimates.

7. Conclusions

Every integer matrix between two Selling-reduced cells of any lattice lies in one set of 6096 matrices with entries up to ± 3. For a measured cell the reduced cells within a strain tolerance are finite and enumerated exactly, and below a radius computed for each cell a lookup of those 6096 matrices between the two reduced cells is a complete search. That lookup is the search we recommend: its 3048 proper matrices are fewer than those of a $\{-1, 0, 1\}$ window, and unlike any fixed window it comes with a check, for each reference, of when it is complete. A $\{-1, 0, 1\}$ window between Selling-reduced cells fails when the reference's reduced superbase has a near-zero conorm on d, which reduction produces near a lattice with right angles from a type I Niggli cell or from an arbitrary basis; between Niggli cells the same window found every operator tested, without a stated range. Prices of candidate operators are geodesic distances, so their coset structure holds exactly, and a pseudo-twofold costs twice its distance to the lattice it would make exact, about 5% per degree of Le Page obliquity. In practice: store each reference as its widened closure, or as its reduced cell with its adjacency radius; Selling-reduce each new data set once; keep the proper candidates within a band of a few per cent, at the upper end of the non-isomorphism of the crystal's Bravais type; group them by the Laue group; and read each class's price against the non-isomorphism distribution. For about one PDB entry in ten a pseudo-symmetric alternative is then as close as a second crystal of the same form, though the cell ranks it first far less often, and the intensities have to decide.

Data availability. The implementation is the module `agentsg.cell.strain_closure` of the `agentsg` repository (`github.com/phzwart/agentsg`). Its main functions are `widened_closure`, `reindex` and `indexing_classes`, with `sort_spectrum` for Appendix B. The PDB cell snapshot is on Zenodo (doi: 10.5281/zenodo.22986222). Every number in this paper is reproduced by a test in `manuscript/appendix_z`, by `agentsg/tests/test_strain_closure.py` or by a script in `manuscript/benchmark`; the supporting information lists them by section, table and figure (Table S10), with the versions of the software used.

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Appendix A
Proofs

Terms. Write a lattice vector in any basis, $x = n_1 v_1 + n_2 v_2 + n_3 v_3$ with integer n_i, and record whether each coordinate is even or odd. This sorts the lattice into eight *parity classes*, the classes of $\Lambda/2\Lambda$; two vectors are in the same class exactly when their difference is twice a lattice vector, so the classes do not depend on the basis. One class is the doubled vectors; the other seven are used here. A *partial sum* of a superbase is the sum of some of its four vectors; there are seven up to sign, one in each of the seven classes. A superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class, that is, a Voronoi vector of the lattice (Voronoi, 1908; Conway & Sloane, 1992; Kurlin, 2022); at a zero conorm the partial sums are Voronoi vectors that are not strict. The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase F in class c is the squared length of its partial sum in that class minus the squared length μ_c of the shortest vector of the class. A partial sum is written $w = \sum s_i v_i$ with $s \in \{0,1\}^4$; a vector of the same class has coordinates $s + 2k$ with k integer. Every

lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{6}$$

whatever the signs of the conorms, because the four vectors sum to zero.

Lemma A1 (window). Let F_0 and F be obtuse superbases of the same lattice. Then every vector of F has superbase coordinates $s_i + 2k_i$ relative to F_0, with $s \in \{0,1\}^4$, $k_0 = 0$ and $|k_i| \le 3$. *Proof.* The non-zero conorms of F_0 connect its four vectors, since otherwise a partial sum would have length zero by equation (6). Every partial sum of F is a shortest vector of its parity class, as is the partial sum of F_0 in that class, so their excess vanishes; by Lemma A2 the excess is $4\sum p^0_{ij}\gamma_{ij}$ with every term non-negative, so $\gamma_{ij} = 0$ on every non-zero slot, i.e. $|k_i - k_j| \le 1$ along a connected graph on four vertices, and with $k_0 = 0$ every $|k_i| \le 3$. □ Enumerating, over all 38 faces, the superbases of such vectors whose six pairwise products are non-positive on the whole face gives 1, 2, 3, 6 and 16 superbases for no zero, one zero, two disjoint zeros, two zeros sharing a vector and three zeros (both placements), independently of Kurlin's lists. In the basis of F_0 their entries are in $\{-1, 0, 1\}$ when the zero conorms avoid d, and up to 3 otherwise (Table 6): the conorms $(0, 0, 3, 2, 0, 5)$ have the obtuse basis $(-3a - 2b - c,\ a,\ a + b)$. Every member lies in $\mathcal{U}$ in each of its 48 labellings, which is the computer-assisted proof that $\mathcal{U}$ contains every arrow, and every member has a labelling with entries in $[-2, 2]$. Relabelling F_0 as well, the largest entry over all members is 1 on every face but the path of three zeros, where it is 2.

Lemma A2 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ a partial sum and w' a vector of its parity class with coordinates $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + \Delta s)$, $\Delta k = k_i - k_j$, $\Delta s = s_i - s_j$; each γ_{ij} is a non-negative integer, zero exactly when $\Delta k \in \{0, -\Delta s\}$. *Proof.* Equation (6), $(\Delta s + 2\Delta k)^2 - \Delta s^2 = 4\Delta k(\Delta k + \Delta s)$, and $t(t + u) \ge 0$ for integers t and $u \in \{-1, 0, 1\}$. □

Lemma A3 (generation). Every arrow of the closure, and every symmetry, is a word

of flips at zero conorms followed by a swap. *Proof.* Let F and F' be obtuse superbases of a lattice with metric $\mathbf{G}$. Add a small positive amount, in the frame of F, to each conorm of F that vanishes under $\mathbf{G}$; this gives a metric $\mathbf{G}'$ for which F is strictly obtuse and hence the only obtuse superbase, and, the lattice having finitely many obtuse superbases, the amount can be chosen so small that every conorm that is positive under $\mathbf{G}$ for any of them stays positive under $\mathbf{G}'$. Selling reduction under $\mathbf{G}'$ started from F' flips only at negative conorms. The first such conorm is zero under $\mathbf{G}$, so the flip carries F' to another obtuse superbase of $\mathbf{G}$, and the same holds at every later step. Reduction terminates (Selling, 1874; Conway & Sloane, 1992), at the only obtuse superbase of $\mathbf{G}'$, which is F. So F' is joined to F by flips at zero conorms of $\mathbf{G}$, and a swap matches the labellings. A symmetry maps F_0 to an obtuse superbase of the lattice; as a matrix it is that flip word followed by a swap. □ Kurlin (2022) lists the superbases of each Voronoi type, and the enumeration of Lemma A1 reproduces the counts.

Proposition A4 (the strain norm bounds every dot product). If $\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F \leq \epsilon$, then $|u^{\mathsf{T}}\Delta\mathbf{G}\,v| \leq \epsilon\,|u|\,|v|$ for all vectors u, v, lengths in $\mathbf{G}$; in particular every conorm of every superbase changes by at most $\epsilon|u_i||u_j|$. *Proof.* With $X = \mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}$, $u^{\mathsf{T}}\Delta\mathbf{G}\,v = (\mathbf{G}^{1/2}u)^{\mathsf{T}}X(\mathbf{G}^{1/2}v)$, at most $\|X\|_2|u||v| \leq \|X\|_F|u||v|$. □

Definition and enumeration A5 (sorts as closed sets of equalities). Fix a zero pattern Z, let O_Z be the arrows of its closure and, for $\mathbf{W} \in O_Z$, let $\mathbf{A}_\mathbf{W}$ be its linear action on conorm vectors (supporting information, Section S3), which on the face $\{p \geq 0,\ p_Z = 0\}$ is a permutation of the slots. For a subspace V of the face put $\mathrm{Sym}(V) = \{\mathbf{W} \in O_Z : \mathbf{A}_\mathbf{W} = \mathrm{id} \text{ on } V\}$ and $\mathrm{Fix}(H) = \{p : \mathbf{A}_\mathbf{W}p = p\ \forall\,\mathbf{W} \in H\}$. A lattice with conorm vector p on the face has point group $\mathrm{Sym}(\{p\})$, every symmetry being in O_Z by Lemma A3. We take as the definition of a sort a maximal set of lattices with the same kind of closure and the same symmetry operators; then the sorts of kind Z are the generic

points of the closed subspaces $V = \mathrm{Fix}(\mathrm{Sym}(V))$ that meet the interior of the face, modulo the arrows that map the face onto itself. This is the classical construction, the fixed subspace of a point group in the space of forms (Delone *et al.*, 1975), written in conorm coordinates; SELLA uses the same subspaces. Sym and Fix are a Galois connection, so the closed subspaces are the intersections of the finitely many subspaces $\mathrm{Fix}(\mathbf{W})$; since $\mathbf{A}_\mathbf{W}$ permutes the slots on the face, each is a set of equalities among conorms. Enumerating the intersections gives 8, 5, 3, 5 and 3 closed subspaces for no zero, one zero, two sharing a vector, two disjoint and three zeros, with symmetry orders $2, 4, 4, 8, 8, 12, 16, 48$; $2, 4, 4, 8, 16$; $4, 8, 24$; $2, 4, 8, 12, 48$; $8, 16, 48$. A point of a closed subspace that lies in the interior of the face has exactly the symmetries $\mathrm{Sym}(V)$; points on its boundary belong to another zero pattern. Matched pattern by pattern, by computation, these are Delone's 24 sorts (supporting information, Table S9).

Lemma A6 (the widened set is finite). If F is within strain $\epsilon < 1$ of obtuse for $\mathbf{G}$, every vector u of F satisfies $|u|^2_\mathbf{G} \le \frac{1+\epsilon}{1-\epsilon}\,\mu_c(\mathbf{G})$, with μ_c the squared length of the shortest vector of the parity class of u, which is that of F_0's partial sum in the class. *Proof.* Let $\mathbf{G}'$ within ϵ make F obtuse. By Proposition A4 with $u = v$, $(1-\epsilon)|u|^2_\mathbf{G} \le |u|^2_{\mathbf{G}'} \le (1+\epsilon)|u|^2_\mathbf{G}$ for every lattice vector. The vectors of F are shortest in their classes for $\mathbf{G}'$, and those of F_0 for $\mathbf{G}$, so $|u|^2_{\mathbf{G}'} \le |v^0_c|^2_{\mathbf{G}'} \le (1+\epsilon)\mu_c(\mathbf{G})$ and $|u|^2_\mathbf{G} \le |u|^2_{\mathbf{G}'}/(1-\epsilon)$. □ The lattice vectors in these seven balls are finitely many and computable from $\mathbf{G}$, so $\mathcal{S}_\epsilon(\mathbf{G})$ is enumerated exactly.

Remark A7 (layers, and the lookup of $\mathcal{U}$). Two superbases are *adjacent* when some positive definite metric makes both obtuse; a superbase adjacent to F_0 is $\mathbf{W}F_0$ with $\mathbf{W} \in \mathcal{U}$, since both are superbases of that metric's closure. Let $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$, and let $r_k(\mathbf{G})$ be the smallest strain from $\mathbf{G}$ to a metric for which a superbase outside $\mathcal{N}_k$ is obtuse. If $\epsilon < r_k(\mathbf{G})$, then $\mathcal{S}_\epsilon(\mathbf{G})$ is the set of members of $\mathcal{N}_k$ within strain ϵ of obtuse; this is the definition of r_k, and its content is the description of $\mathcal{N}_1$,

which rests on the computer-assisted completeness of $\mathcal{U}$ (Lemma A1). *The lookup.* Let $\mathbf{P}$ be an operator of price at most ϵ from a reference with reduced basis B_R to a new cell with reduced basis B_N, and let $e^{\epsilon}-1 < r_1$ for the reference. By Lemma A8 the new cell's reduced superbase, carried back by $\mathbf{P}$, lies in $\mathcal{S}_{e^{\epsilon}-1}(\mathbf{G}_R) \subseteq \mathcal{N}_1$, so it is $\mathbf{W}F_0$ for some $\mathbf{W} \in \mathcal{U}$ in some labelling, and $\mathbf{P} = B_R\mathbf{W}\mathbf{Q}B_N^{-1}$ with $\mathbf{Q}$ a swap. Since $\mathcal{U}$ is closed under swaps, $\mathbf{W}\mathbf{Q} \in \mathcal{U}$: trying $B_R\mathbf{W}B_N^{-1}$ for the 6096 matrices $\mathbf{W}$ finds $\mathbf{P}$. □ The radii are computed exactly from the balls of Lemma A6 at a strain of $\frac{1}{4}$. No lower bound on r_1 in terms of the cell's geometry is proved; for the cells of Table 7 below the cap it is of the order of the ratio of the shortest to the longest edge.

Lemma A8 (widening one side suffices). (i) If $\Delta(\mathbf{A},\mathbf{B}) \le \epsilon$, then $\|\mathbf{B}-\mathbf{A}\|_{\mathbf{A}} \le e^{\epsilon}-1$. (ii) Let $\mathbf{P}$ be an operator from a reference with metric $\mathbf{G}_R$ to a new cell with metric $\mathbf{G}_N$, of price at most ϵ, and let F be obtuse for $\mathbf{G}_N$. Then $\mathbf{P}F \in \mathcal{S}_{e^{\epsilon}-1}(\mathbf{G}_R)$. *Proof.* (i) The eigenvalues of $\mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2}$ are e^{l_i} with $\sum l_i^2 \le \epsilon^2$, and $|e^{l}-1| \le |l|(e^{a}-1)/a$ for $|l| \le a$, because $(e^{x}-1)/x$ increases; with $a = \max|l_i| \le \epsilon$, $\|\mathbf{B}-\mathbf{A}\|_{\mathbf{A}} = (\sum(e^{l_i}-1)^2)^{1/2} \le \epsilon\,(e^{\epsilon}-1)/\epsilon$. (ii) The price is $\Delta(\mathbf{G}_N, s\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P})$ for the fitted scale s. Carried back by $\mathbf{P}$, an isometry of both measures, $\mathbf{P}F$ is obtuse for $\mathbf{M} = \mathbf{P}^{-\mathsf{T}}\mathbf{G}_N\mathbf{P}^{-1}$, and $\Delta(s\mathbf{G}_R, \mathbf{M}) \le \epsilon$, so by (i) $\mathbf{M}$ is within strain $e^{\epsilon}-1$ of $s\mathbf{G}_R$; the strain and the widened set are unchanged by the scale.□ For the price band of 7.4% the reference is widened at 7.7%.

Proposition A9 (the price of a symmetry). Let g be an integer matrix of finite order n and $V_g = \{\mathbf{G} : g^{\mathsf{T}}\mathbf{G}g = \mathbf{G}\}$ the metrics for which g is a symmetry. (i) $\Delta(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g) \le 2\Delta(\mathbf{G}, V_g)$, with equality when g is a twofold. (ii) If g is a twofold along a lattice row of obliquity δ in $\mathbf{G}$ (the angle between the row and the normal to the lattice plane that g negates), then $\Delta(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g) = 2\sqrt{2}\,\operatorname{artanh}(\sin\delta)$. (iii) For a metric $\mathbf{G}^* \in V_g$ and the strain at $\mathbf{G}^*$, $\|g^{\mathsf{T}}\mathbf{G}g - \mathbf{G}\|_{\mathbf{G}^*}$ lies between $\rho_{\min}$ and $\rho_{\max}$ times the distance from $\mathbf{G}$ to V_g in that norm, with $\rho = |e^{i\theta}-1|$ over the rotation

angles $\theta \neq 0$ of $X \mapsto g^{\mathsf{T}} X g$: $\rho = 2$ for a twofold, $\sqrt{3}$ for a threefold, $\{\sqrt{2}, 2\}$ for a fourfold and $\{1, \sqrt{3}\}$ for a sixfold. The scale fit leaves these prices unchanged, since $\det g^{\mathsf{T}} \mathbf{G} g = \det \mathbf{G}$. *Proof.* (i) $\mathbf{G} \mapsto g^{\mathsf{T}} \mathbf{G} g$ is an isometry of Δ. For any $\mathbf{G}' \in V_g$, $\Delta(\mathbf{G}, g^{\mathsf{T}} \mathbf{G} g) \leq \Delta(\mathbf{G}, \mathbf{G}') + \Delta(\mathbf{G}', g^{\mathsf{T}} \mathbf{G} g) = 2\Delta(\mathbf{G}, \mathbf{G}')$. For a twofold, the geodesic midpoint m of $\mathbf{G}$ and $g^{\mathsf{T}} \mathbf{G} g$ is unique and g exchanges the two ends, so $m \in V_g$ and $\Delta(\mathbf{G}, V_g) \leq \frac{1}{2}\Delta(\mathbf{G}, g^{\mathsf{T}} \mathbf{G} g)$. (ii) In Cartesian coordinates for $\mathbf{G}$, g becomes $T = \mathbf{G}^{1/2} g \mathbf{G}^{-1/2}$, an involution fixing the line ℓ of the row and negating the plane Π; it is $2\mathbf{P} - I$ with $\mathbf{P}$ the projection onto ℓ along Π, and the angle between ℓ and the normal of Π is δ. The price is $\| \ln(T^{\mathsf{T}} T) \|_F$. $T^{\mathsf{T}} T$ is the identity on the line of Π normal to ℓ, and on the plane spanned by ℓ and the normal of Π it has eigenvalues $(1 \pm \sin\delta)/(1 \mp \sin\delta)$, whose logarithms are $\pm 2 \operatorname{artanh}(\sin\delta)$. (iii) $\sigma : X \mapsto g^{\mathsf{T}} X g$ is an isometry of the inner product $\operatorname{tr}(\mathbf{G}^{*-1} X \mathbf{G}^{*-1} Y)$ of order n, so the space of symmetric matrices splits orthogonally into its fixed space, which is $V_g - \mathbf{G}^*$, and planes and lines on which it rotates by $\theta = 2\pi k/n$. With $\mathbf{G} - \mathbf{G}^* = X_0 + X_\perp$, X_0 fixed, the distance to V_g is $\|X_\perp\|$ and $g^{\mathsf{T}} \mathbf{G} g - \mathbf{G} = (\sigma - 1) X_\perp$. The rotation angles are those of the products $\lambda_i \lambda_j$, $i \leq j$, of the eigenvalues of g, squares included; the squares give the angle π of a fourfold and $2\pi/3$ of a sixfold. □ Part (iii) is exact for the strain at $\mathbf{G}^*$; for the price it holds to first order in the distance to V_g. No exact relation to Le Page's obliquity holds for orders 3, 4 and 6: at Le Page's boundary the fourfolds of a body-centred cubic lattice cost a median 15.0%, more than its twofolds (Table B1). The metrics of a sort to which g belongs lie inside V_g (read in the frame in which the sort's distance is attained), so, to first order in the error, where the strain of a sort distance and the geodesic distance agree, a twofold's price is at most twice the distance to that sort, with equality when V_g is the sort's subspace, as for the twofold of a primitive monoclinic lattice.

Proposition A10 (candidates and cosets). Let $\mathbf{P}_0$ be the true operator from a ref-

erence R to a new cell N, of price p_0; let H_ϵ be the proper unimodular h with $\pi(h) = \Delta(\mathbf{G}_R, h^\mathsf{T}\mathbf{G}_R h) \leq \epsilon$, a finite set; and let C_ϵ be the candidates, the proper unimodular operators of price at most ϵ. Then (i) $\pi(h_1 h_2) \leq \pi(h_1) + \pi(h_2)$; (ii) $H_{\epsilon - p_0}\mathbf{P}_0 \subseteq C_\epsilon \subseteq H_{\epsilon + p_0}\mathbf{P}_0$; (iii) if no self-map of R has price in $(\epsilon, 2\epsilon]$, H_ϵ is a group; (iv) if none has price in $(\epsilon - p_0, \epsilon + p_0]$, then $C_\epsilon = H_\epsilon \mathbf{P}_0$. *Proof.* The price is the geodesic distance between metrics normalised to unit determinant, and unimodular matrices act on those by isometries. (i) $\Delta(\mathbf{G}_R, (h_1 h_2)^\mathsf{T}\mathbf{G}_R h_1 h_2) \leq \Delta(\mathbf{G}_R, h_2^\mathsf{T}\mathbf{G}_R h_2) + \Delta(h_2^\mathsf{T}\mathbf{G}_R h_2, h_2^\mathsf{T} h_1^\mathsf{T}\mathbf{G}_R h_1 h_2) = \pi(h_2) + \pi(h_1)$. (ii) By the triangle inequality the price of $h\mathbf{P}_0$ differs from $\Delta(\mathbf{P}_0^\mathsf{T}\mathbf{G}_R\mathbf{P}_0, \mathbf{P}_0^\mathsf{T} h^\mathsf{T}\mathbf{G}_R h\mathbf{P}_0) = \pi(h)$ by at most p_0, and every unimodular $\mathbf{P}$ is $h\mathbf{P}_0$ with $h = \mathbf{P}\mathbf{P}_0^{-1}$. (iii) By (i) a product of two elements of H_ϵ has price at most 2ϵ, hence at most ϵ; a finite set of matrices closed under products is a group. (iv) The two sets of (ii) then coincide. □ The conditions are sufficient, not necessary: the worked cell at $\epsilon = 7.4\%$ has self-maps of price 0, 1.55% and 10.2%, so the condition of (iii) fails, yet $H_\epsilon = \{\mathbf{I}, 2_y\}$ is a group, which the implementation checks directly.

Appendix B
Distances to the 24 sorts

The same computation gives the distance from a measured cell to each of the 24 sorts. In each of the 6096 frames of $\mathcal{U}$ the cell's conorms are projected, in the strain, onto the sort's subspace (its zeros and equalities, Definition A5, the other conorms non-negative), and the smallest distance is taken. This is the computation of SELLA (Andrews *et al.*, 2023), which projects the boundary images of a cell in S^6 onto the same subspaces, and of Oishi-Tomiyasu (2012) in Selling coordinates; projection onto Bravais-lattice subspaces goes back to Andrews & Bernstein (1988). What changes is

the norm, which is the same in every frame, and the set of frames, which is $\mathcal{U}$. We call the 24 distances the *sort spectrum* of the cell. A distance below the adjacency radius r_1 is exact, since the frames of $\mathcal{U}$ then contain the frame of the nearest lattice of the sort; a minimum above r_1 is an upper bound over the frames used, the true distance being at least r_1, and the implementation flags it (`certified_sort_spectrum`). Along every specialisation of Table S2 the distance cannot decrease, since the lower sort's subspace lies inside the upper one's.

Fig. B1 draws the spectrum of the worked cell on the graph of specialisations. The nearest sorts are A2 at 0.28%, M4 (mP) at 0.78%, A3 at 3.8% and O5 (oP) at 5.1%, and every other sort is at least 25% away, a bound that holds because r_1 reaches the cap for this cell. For a tetragonal lysozyme cell indexed in $P1$ the twelve sorts from aP to tP all lie within 0.22% (supporting information, Fig. S1).

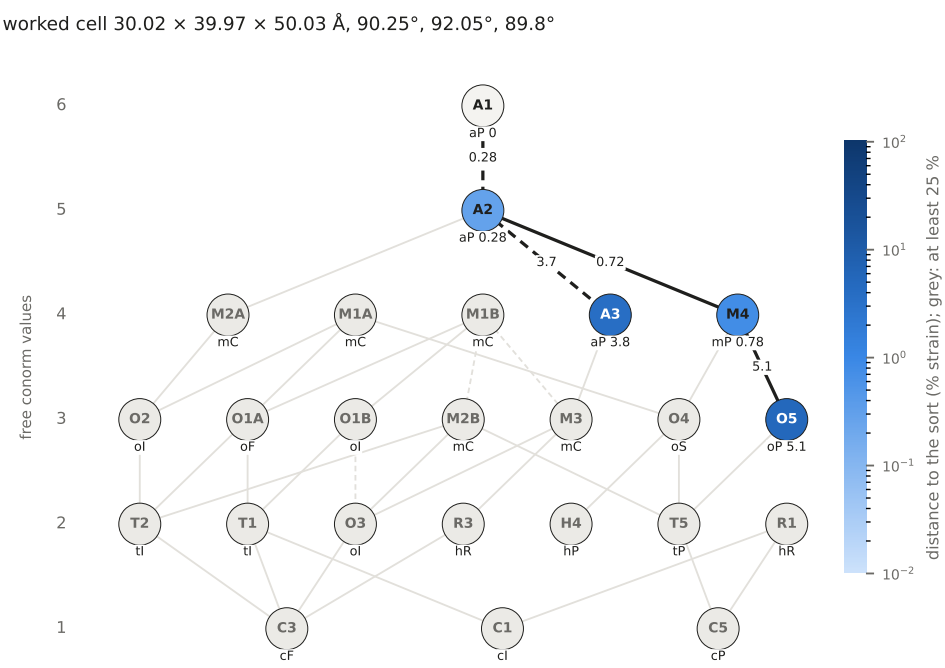


Fig. B1. The 24 Delone sorts as a graph for the measured worked cell. Rows are the number of free conorm values. Each node gives the Bravais type and the strain distance (%) from the cell to the sort, its colour the same distance on a log scale; sorts at least 25% away are grey. Edges are the specialisations of Table S2, labelled with the distance from the cell idealised onto the upper sort to the lower one. Dark edges are routes on which these steps add in quadrature to the lower sort's distance within 3%, drawn where that distance is below 10%. Dashed edges keep the Bravais type.

Le Page's 3° *by sort.* The distance to a sort at Le Page's boundary depends on the sort, while the price of each operator does not. For exact lattices of each sort perturbed in random directions until Le Page's obliquity for the sort's Bravais group is exactly 3°, the median largest twofold price is 14.8% in every sort, which Proposition A9 makes true by construction and the table confirms numerically (larger only in the monoclinic tails, up to 40–55%, where Le Page's group is another C-centred group), while the median distance to the sort is 7.4–7.7% for the orthorhombic and monoclinic sorts other than O3, M2B and M3, 7.9–8.1% for tP, tI, hP and hR, and 8.3–8.5% for the cubic sorts, with longer tails for the boundary sorts O3, M2B and M3 (Table B1).

A single band on the distance to a sort therefore tests each sort at a different level, which is why the reindexing in this paper is decided on the prices of operators.

Table B1. *Le Page's* 3° *by Delone sort: for exact lattices of each sort, each perturbed in a random direction until the obliquity of the sort's Bravais group is* 3°*, the strain distance to the sort (median, and 10th–90th percentiles) and the median of the largest price of the lattice's twofolds and of its other rotations, by order, in %. In the monoclinic sorts some perturbations reach the boundary through a different C-centred group, whose own twofold costs* 14.8% *while the exact lattice's costs more. Of 25 directions per sort, those that never reach an obliquity of* 3° *are dropped, hence* $n < 25$ *in seven sorts; with* $n \leq 25$ *the percentiles are indicative.*

sort	type	n	distance to sort	twofold price	other orders
C1	cI	25	8.3 (7.6–9.0)	14.8	3: 14.1, 4: 15.0
C3	cF	25	8.3 (7.7–9.2)	14.8	3: 14.3, 4: 14.5
C5	cP	25	8.5 (7.6–9.3)	14.8	3: 14.2, 4: 14.7
H4	hP	25	7.9 (7.5–8.5)	14.8	3: 13.7, 6: 11.3
T1	tI	25	7.9 (7.5–9.1)	14.8	4: 14.0
T2	tI	25	8.0 (7.6–8.8)	14.8	4: 14.4
T5	tP	25	8.0 (7.4–8.9)	14.8	4: 14.1
R1	hR	25	8.0 (7.5–8.6)	14.8	3: 14.4
R3	hR	25	8.1 (7.5–8.8)	14.8	3: 14.5
O1A	oF	24	7.5 (7.4–7.8)	14.8	—
O1B	oI	25	7.6 (7.4–8.1)	14.8	—
O2	oI	25	7.6 (7.4–8.2)	14.8	—
O3	oI	25	8.6 (7.6–13.6)	14.8	—
O4	oS	25	7.7 (7.4–8.2)	14.8	—
O5	oP	25	7.5 (7.4–8.3)	14.8	—
M1A	mC	20	7.4 (7.4–9.1)	14.8	—
M1B	mC	15	7.4 (7.4–7.4)	14.8	—
M2A	mC	18	7.4 (7.4–8.5)	14.8	—
M2B	mC	17	9.3 (7.5–19.9)	14.8	—
M3	mC	23	8.9 (7.5–16.1)	14.8	—
M4	mP	23	7.4 (7.4–7.4)	14.8	—

Against BLDConograph and SELLA. On the seven sample cells of BLDConograph (Oishi-Tomiyasu, 2024) the highest Bravais type within 7.4% is the program's type in all seven. Against SELLA on fourteen cells the rankings of the 24 sorts agree closely (Spearman's rank correlation 0.77 to 0.97), while the two scales are not proportional and differ most on needles and plates, where an absolute band weighs the long edges differently (supporting information, Section S5 and Table S5).

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Synopsis

Every integer matrix between two Selling-reduced cells of any lattice lies in one set of 6096 matrices, and for a measured cell a lookup of that set is a complete reindexing search below a radius computed for each cell. Pseudo-symmetric alternatives are priced and compared with the non-isomorphism of PDB crystals, which says when the cell alone cannot exclude them.